

AI Agent Foundations and Evolution

Origins of Intelligent Systems

Artificial intelligence evolved from symbolic systems toward learning-based systems. Early AI systems relied on explicit logic and rules, but modern systems increasingly depend on neural architectures. A central turning point emerged through attention mechanisms and transformer architectures. Researchers discovered that systems could model relationships across large sequences and preserve context across tasks.

The evolution of AI agents created a distinction between models and agents. A model predicts outputs. An agent observes environments, stores state, plans actions, and executes tool usage. This distinction became increasingly important with large language models and autonomous workflows.

Repeated concepts: transformers, attention, memory systems, tool use, retrieval augmented generation, computer use, and planning. Artificial intelligence evolved from symbolic systems toward learning-based systems. Early AI systems relied on explicit logic and rules, but modern systems increasingly depend on neural architectures. A central turning point emerged through attention mechanisms and transformer architectures. Researchers discovered that systems could model relationships across large sequences and preserve context across tasks.

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Memory and Tool Use

Memory transformed agents from one-step systems into persistent systems. Long-term memory architectures, retrieval systems, episodic memory, semantic memory, and external knowledge stores became important.

Tool use emerged as another capability. Agents increasingly interacted with browsers, APIs, spreadsheets, and databases. Tool execution introduced planning loops: observe, reason, act, verify, and repeat.

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